MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1MDCSTA100 – Statistical Data Collection

Duration: 1Hour Maximum Marks: 35

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

*Answer **All** Questions.*

*Each question carries **2** marks*

1. What is the primary goal of an experimental study?
   a) To establish causal relationships between variables b) To gather opinions and demographic information c) To explore naturally occurring events
   d) To provide open-ended insights into human behaviour. [Remember] [CO1]
2. Which of the following features is characteristic of quantitative research?
   a) Objective measurement b) Statistical analysis
   c) Structured methods d) All of the above. [Apply] [CO5]
3. At what stage of the research process is the hypothesis typically formulated?
   a) Data collection b) Data analysis
   c) Literature review d) Research design [Understand] [CO1]
4. Lottery method is an example of
   a) Non probability sampling b) Cluster sampling
   c) Simple Random Sampling d) None of these. [Remember] [CO2]
5. Daily rainfall is an example of what sort of data:
   a) Discrete b) Continuous c) Both discrete and continuous
   d) Neither discrete nor continuous. [Understand] [CO1]
6. Which scale is the simplest form of measurement?
   a) Nominal b) Ordinal c) Interval d) Ratio. [Understand] [CO3]
7. In the context of experimental design, what does randomization help to eliminate?
   a) Systematic error b) Data variability c) Human error
   d) Natural variation [Analyse] [CO7]
8. What type of design is most commonly used in agricultural research?
   a) CRD b) RBD c) LSD d) Factorial design [Apply] [CO7]
9. In practice, how are the treatments assigned in a Latin Square Design?
   a) Randomly within each row and column b) Sequentially c) Based on previous results
   d) Fixed to specific rows and columns [Understand] [CO7]
10. Which of the following is a benefit of using factorial experiments?
    a) Reduced experimental error b) Increased precision c) Improved understanding of interaction of effects d) All of these [Remember] [CO7] **(2 x 10 = 20)**

Part B

Short Essay Type questions

*Answer any **3** questions.*

*Each question carries **5** marks.*

11. Define extraneous variable. Give two examples of extraneous variables. [Remember] [CO1]
12. What are the advantages of sampling over complete enumeration? [Understand] [CO2]
13. Explain Quantitative and Qualitative data. [Understand] [CO1]
14. Why should questions in a questionnaire be clear and concise? [Apply] [CO6]
15. Discuss the main advantages of RBD. [Analyse] [CO7]

(5 x 3=15)